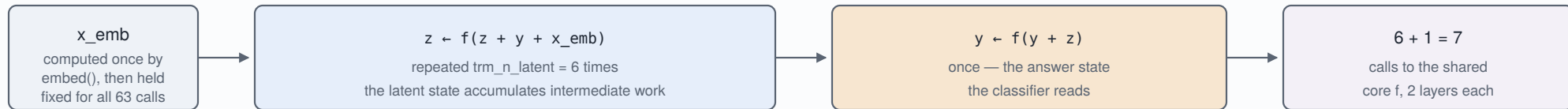


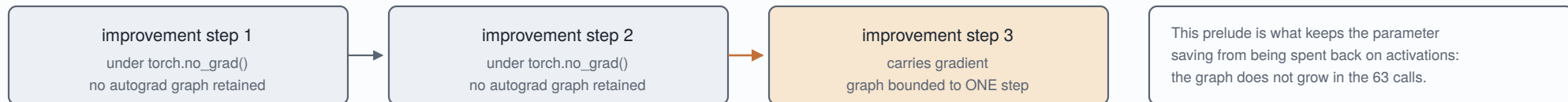
The recursion schedule: 63 applications of one two-layer core per forward pass

Read off `RecursiveCore.improve` / `one_segment` and `GMedMambaRecursive.forward_deep_supervision`. The run configuration is `n_latent 6`, `n_improve 3`, `segments 3`.

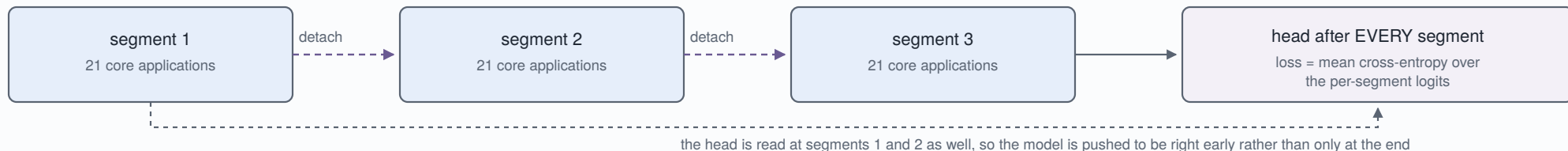
One improvement step — `RecursiveCore.improve` — 7 core applications



One segment — `RecursiveCore.one_segment` — 3 improvement steps, 21 core applications



One forward pass — deep supervision over 3 segments, 63 core applications



$(6 + 1) \times 3 \times 3 = 63$ applications of the same 2-layer core, so 126 mixer calls — and **`forward()` runs the whole schedule at inference too.**

Every core application is gradient-checkpointed (`trm_checkpoint_core`). Without it the model exhausts 16 GB of VRAM on the hyperspectral arm.